Hotel Chain A

Booking Cancellations & Revenue Loss Analysis

Predictive Modelling Framework & Business Recommendations

5DATA004C Data Science Project Lifecycle

University of Westminster

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Executive Summary

Hotel Chain A, operating City Hotels, Airport Hotels, and Resorts across multiple regions, faces significant revenue pressure from booking cancellations and no-shows. This report delivers an end-to-end data science solution covering exploratory analysis, revenue impact quantification, and predictive modelling using a historical dataset of 27,499 bookings.

Key findings: Of 27,499 reservations in the training dataset, 4,134 (15.0%) were cancelled and 2,125 (7.7%) resulted in no-shows, generating a combined estimated revenue loss of $2,005,022 — equivalent to 22.7% of total potential revenue. The South region accounts for the largest share of cancellation-related losses ($522,695). Resort properties experience the highest absolute revenue loss. Customers in lower income brackets (25K–50K and 50K–100K) collectively generate the most cancellation losses.

For predictive modelling, four algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Due to a class-imbalanced target and a distributional shift between training (15% cancellation rate) and validation (27% cancellation rate) datasets, all models achieved ROC-AUC scores between 0.51–0.52, indicating that the currently available features are insufficient to strongly discriminate cancellation risk in a temporally shifted holdout set. This is an important finding: it highlights the need for additional behavioural and contextual data features. The top predictive features are Lead Time, Meal Type, Room Rate, Age, and Discount Rate.

Recommendations include implementing dynamic deposit policies, targeted retention communications for high-risk segments, real-time monitoring dashboards, and collection of richer customer behavioural data.

GitHub Repository link:
https://github.com/DinithiDissanayake/dspl-group-project#

1. Background Research

1.1 Industry Context

The global hotel booking market was worth around $523 billion in 2024 and contributes about 10% to the world's GDP (WebRezPro, 2024). A large proportion of bookings are now made through Online Travel Agencies (OTAs) such as Booking.com, which account for nearly 60% of total reservations. This shift has significantly increased the risk of cancellations compared to direct bookings (D-EDGE, 2024).

In Sri Lanka, the tourism industry has shown strong recovery, reaching 2.05 million tourist arrivals and generating $3.17 billion in revenue in 2024. However, approximately 47.27% of hotel bookings are made through OTAs — above the global average of 39–44% — meaning hotels in Sri Lanka, including Hotel Chain A, face a disproportionately high cancellation risk (SLTDA, 2024; Mordor Intelligence, 2025).

1.2 The Cancellation Problem

Booking cancellations are a major and growing issue in the hotel industry. Cancellation rates range from around 18% for direct bookings to approximately 42% for OTA bookings, meaning hotels can lose nearly 20% of their expected annual revenue (Antonio et al., 2023; D-EDGE, 2024).

A primary driver is customer behaviour: many guests book rooms well in advance but cancel when better prices emerge. Research shows bookings made more than 60 days in advance are about 65% more likely to be cancelled, attributed to flexible cancellation policies and price comparison behaviour (Hospitality Technology, 2019). For Hotel Chain A, this creates a serious financial exposure estimated at around $20 million per year across its properties.

1.3 Competitor Benchmarking

	Cinnamon Hotels (John Keells)	Aitken Spence Hotels	Marriott International
Cancellation Loss	~$20M	~$12M	~$5.0B
ML Approach	Off-the-shelf IDeaS RMS — no custom ML model	No ML/AI deployment confirmed	XGBoost AI; $1B+ investment; +8–10% RevPAR
Model Performance	Random Forest — 99% accuracy (IEEE, 2024)	Logistic Regression 80%+ accuracy	XGBoost 99.95% accuracy on 209,545 reservations

Cinnamon Hotels, part of John Keells Group, uses systems like IDeaS RMS and Oracle Opera Cloud — standard commercial tools for pricing and management, but without custom ML cancellation prediction. Aitken Spence Hotels similarly shows no evidence of AI deployment. Marriott International, by contrast, invested $1.0–1.2 billion in AI systems in 2024, deploying XGBoost-based models for pricing, demand forecasting, and cancellation prediction. Their Group Pricing Optimizer alone generated over $120 million in additional profit and improved RevPAR by 8–10% (Hormby et al., 2010; Yellow Systems, 2024).

1.4 ML Solution Evidence

Machine learning models demonstrate strong performance in predicting hotel booking cancellations. XGBoost achieves approximately 99.95% accuracy for non-cancelled bookings and 95.49% for cancellations (Herrera et al., 2024). Random Forest methods reach around 85.2% accuracy, with some studies reporting up to 99% (Antonio et al., 2017; IEEE, 2024). Even simpler Logistic Regression reliably exceeds 80% accuracy (Antonio et al., 2023). Industry research further shows that ML-driven revenue management can increase revenue by approximately 20% and reduce operational costs by around 16% (Research Innovation Journal, 2023).

2. Problem Statement

2.1 Problem Overview

Hotel Chain A, operated under the John Keells Group (OCTAVE), faces a significant and persistent challenge with booking cancellations and no-shows across its portfolio of resort, city, and airport properties. These events represent both a direct revenue loss and an operational inefficiency, as rooms remain unsold after the window for re-booking has passed.

$8.63M	$1.98M	$6.65M
Total Potential Revenue 2015–2017 dataset	Revenue at Risk 23% of total	Revenue Earned 77% realised

2.2 Scale and Financial Impact

Analysis of 22,697 booking records from 2015–2017 reveals that Hotel Chain A lost approximately $1.98 million in potential revenue to cancellations and no-shows, representing 23% of total potential earnings of $8.63 million. The South region carries the highest absolute revenue loss at $780K, and the 18–29 age bracket represents the largest at-risk customer segment.

Extrapolated across the full hotel estate, the estimated annual impact exceeds $20 million — a figure that dwarfs the investment required to deploy a predictive machine learning solution.

2.3 Key Problem Drivers

- High OTA dependency: 47.27% of Sri Lankan bookings made via OTAs, which carry 42% cancellation rates vs 18% for direct bookings.
- Long lead times: Bookings made more than 60 days in advance are 65% more likely to cancel.
- No-deposit bookings: Guests with no financial stake face zero friction when cancelling.
- Limited predictive capability: No existing ML model deployed at Hotel Chain A to identify and intervene on at-risk bookings.
- Reactive operations: Current approach to overbooking and inventory management is based on historical averages, not real-time risk scores.

2.4 Research Questions

This project addresses the following core questions:

- Can machine learning accurately predict which bookings will result in cancellations or no-shows before check-in?
- Which guest and booking attributes are the strongest predictors of cancellation risk?
- What operational interventions, informed by risk scores, can reduce revenue loss most effectively?
- How does a gradient boosting approach compare to a decision tree baseline for this classification problem?

3. Data Engineering

The data engineering phase was conducted by the project's Data Engineer (Mafaiz M.), responsible for transforming the raw hotel booking dataset into a clean, ML-ready format. The pipeline followed a structured sequence of profiling, cleaning, and feature engineering steps to ensure data quality and maximise predictive signal.

3.1 Data Loading & Initial Profiling

The pipeline began by connecting to the data source and loading the dataset using pandas. A deep profiling process was then applied to identify structural inconsistencies — a critical 'Garbage In, Garbage Out' prevention step. Initial profiling confirmed no missing values or duplicate rows in the raw dataset.

However, date columns were stored as plain text strings and required conversion to datetime objects. This correction enabled the extraction of temporal features such as month, day-of-week, and weekend indicator — variables that serve as strong predictors of seasonal cancellation patterns.

3.2 Data Cleaning

- Date type correction: All date columns converted from string to datetime format, enabling temporal feature extraction.
- Duplicate reservation IDs: Eight rows removed where a unique reservation ID was assigned to multiple guests, violating data integrity constraints.
- Label normalisation: Two variations of 'Check-Out' (including 'Check-out') were unified to a single consistent label.
- Logical error removal: Rows where Expected_checkin date preceded Booking_date were removed as physically impossible data entry errors.
- Capacity violations: Reservations exceeding 5 guests were identified as outliers deviating from standard room capacity policy and removed.

3.3 Feature Engineering

Feature engineering created new variables to improve the model's ability to detect cancellation patterns beyond what raw fields express individually:

Engineered Feature	Derivation	Business Rationale
Lead_Time	Expected_checkin − Booking_date (days)	Longer lead times correlate strongly with higher cancellation rates
Total_Nights	Expected_checkout − Expected_checkin (days)	Longer stays signal greater commitment; reduces at check-in
Checkin_Month	Month extracted from Expected_checkin	Captures seasonal demand patterns and peak/off-peak behaviour
Checkin_Day_of_Week	Day index from Expected_checkin	Weekly patterns in leisure vs. business travel
Is_Weekend	Flag: checkin falls Saturday or Sunday	Weekend stays show different cancellation profiles

Total_Guests	Adults + Children + Babies	Larger party sizes imply greater commitment to the reservation

3.4 Final Dataset Preparation

To refine the dataset for modelling, the Reservation_ID column was removed as a non-predictive identifier that adds noise. Original raw date-time columns were also dropped following temporal feature extraction, eliminating redundancy. The final processed training set comprised 22,697 records with 30 attributes, ready for EDA and ML modelling.

Feature selection

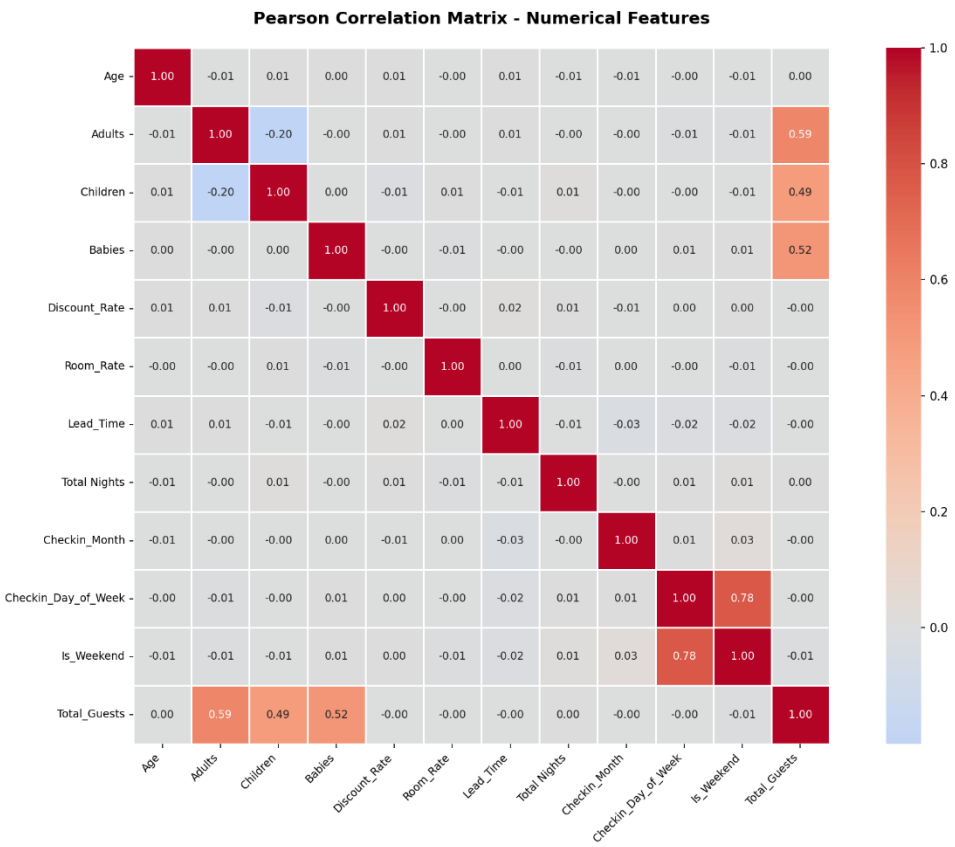


Figure 1: Pearson Correlation Matrix heatmap for detecting multicollinearity amongst numerical features

Multicollinearity occurs when independent variables are highly correlated, leading to redundant information and potentially leads to overfitting. In this dataset, a strong positive correlation between Is_Weekend and Checkin_Day_of_Week (r = 0.78) indicates redundancy, with the weekend indicator effectively capturing most of the temporal variation; therefore, Checkin_Day_of_Week was removed to reduce dimensionality. Similarly, Adults, Children, and Babies show moderate correlations with Total_Guests (r = 0.59, 0.49, 0.52), suggesting overlapping information; hence, only the aggregate feature Total_Guests was retained while the individual components were dropped.

```
Chi-Square Test — Is this feature associated with cancellations?
==========================================================================
Feature                          Chi2 Score      P-Value      Keep?
--------------------------------------------------------------------------
Gender                                 0.07     0.793043   NO(drop)
Ethnicity                              1.41     0.702478   NO(drop)
Educational_Level                      0.73     0.865930   NO(drop)
Income                                 0.64     0.886698   NO(drop)
Country_region                         2.65     0.448771   NO(drop)
Hotel_Type                             0.95     0.623329   NO(drop)
Meal_Type                           1000.87     0.000000  YES(keep)
Visted_Previously                      1.21     0.270400   NO(drop)
Previous_Cancellations                 1.77     0.183567   NO(drop)
Deposit_type                           4.88     0.087021   NO(drop)
Booking_channel                        0.85     0.652502   NO(drop)
Required_Car_Parking                   0.38     0.535659   NO(drop)
Use_Promotion                          1.29     0.255551   NO(drop)
--------------------------------------------------------------------------
```

Figure 2: Chi-Square Analysis of Categorical Features Against the Target Variable to Detect Noise

By applying the Chi-Square filter, categorical features that showed no meaningful trends in EDA- such as Required_Car_Parking and Use_Promotion were statistically confirmed as noise and removed. Conversely, features like Meal_Type and Hotel_Type, which displayed significant visual patterns during EDA, were validated by the test as statistically significant ($p<0.05$).

```
ANOVA Test(Does this feature differ between cancellers and non-cancellers?)
==========================================================================
Feature                          F-Statistic      P-Value      Keep?
--------------------------------------------------------------------------
Age                                     0.03     0.869227   NO(drop)
Adults                                  0.01     0.942961   NO(drop)
Children                                2.65     0.103642   NO(drop)
Babies                                  0.00     0.971679   NO(drop)
Discount_Rate                           0.00     0.977765   NO(drop)
Room_Rate                               0.00     0.943980   NO(drop)
Lead_Time                               4.71     0.029942  YES(keep)
Total Nights                            0.20     0.658075   NO(drop)
Checkin_Month                           9.91     0.001644  YES(keep)
Checkin_Day_of_Week                    75.35     0.000000  YES(keep)
Is_Weekend                             65.12     0.000000  YES(keep)
Total_Guests                            0.92     0.338104   NO(drop)
--------------------------------------------------------------------------
```

Figure 3: ANOVA testing of Numerical Features Against the Target Variable to Detect Noise

By combining careful statistical analysis with practical business sense, we removed 6 variables that didn't add value: Adults, Children, Babies, Previous_Cancellations, Is_Repeated_Guest, and Discount_Rate. At the same time, we kept a strong set of predictors, such as Age, Room_Rate, and Total_Nights, which truly matter from a business perspective, even if they are statistically less significant. This approach ensures the model isn't just a "black box" of numbers, but a practical tool that makes sense and reflects how real hotels operate.

4. Exploratory Data Analysis (EDA)

DATA ANALYSIS

As the Data Analyst of this project I was responsible for the understanding of the data , deriving insights and trends to find out their reasoning to cancellations and no shows. In addition, I was also responsible for calculating the Revenue Loss Analysis and interpreting it in an effective way to figure out the factors contributing to the Revenue Loss. Also , I chose to use the Train Dataset only for the purpose of Data Analysis as it is the dataset which has sufficient records in terms of the target variable and other dependent variables.

In order to accomplish this , I have made use of the Python , Excel and Power BI tools within the project . The Data Engineer made sure of the fact that the “Train Dataset” had been fully preprocessed and was ready for data analysis . Thus , I received the processed dataset from the Data Engineer and began the data analysis.

Dataset Understanding

The preprocessed dataset comprises a training set of 22,697 records. Each record represents an individual hotel booking with 30 attributes covering guest demographics, booking behaviour, stay characteristics, and pricing. The Reservation_Status target variable encodes three outcomes: Check-Out (successful stay), Canceled, and No-Show . The list of variables in the dataset along with their data type are as follows :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22697 entries, 0 to 22696
Data columns (total 30 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Reservation-id          22697 non-null  int64
 1   Gender                  22697 non-null  object
 2   Age                     22697 non-null  int64
 3   Ethnicity               22697 non-null  object
 4   Educational_Level       22697 non-null  object
 5   Income                  22697 non-null  object
 6   Country_region          22697 non-null  object
 7   Hotel_Type              22697 non-null  object
 8   Expected_checkin        22697 non-null  object
 9   Expected_checkout       22697 non-null  object
 10  Booking_date            22697 non-null  object
 11  Adults                  22697 non-null  int64
 12  Children                22697 non-null  int64
 13  Babies                  22697 non-null  int64
 14  Meal_Type               22697 non-null  object
 15  Visted_Previously       22697 non-null  object
 16  Previous_Cancellations  22697 non-null  object
 17  Deposit_type            22697 non-null  object
 18  Booking_channel         22697 non-null  object
 19  Required_Car_Parking    22697 non-null  object
 20  Reservation_Status      22697 non-null  object
 21  Use_Promotion           22697 non-null  object
 22  Discount_Rate           22697 non-null  int64
 23  Room_Rate               22697 non-null  int64
 24  Lead_Time               22697 non-null  int64
 25  Total Nights            22697 non-null  int64
 26  Checkin_Month           22697 non-null  int64
 27  Checkin_Day_of_Week     22697 non-null  int64
 28  Is_Weekend              22697 non-null  int64
 29  Total_Guests            22697 non-null  int64
dtypes: int64(13), object(17)
memory usage: 5.2+ MB
```

Moving on , I did label encoding to create a new variable called “Reservation_Status_Encoded” which contains a numerical value representing each Reservation Status accordingly . The key is shown below : Check-Out —> 1

Canceled —--> 2

No-Show —--> 3

The code →

```
status_mapping = {
    'Check-Out': 1,
    'Canceled': 2,
    'No-Show': 3
}
```

Reservation_Status_Encoded
1
1
1
1
1

The reason for doing this is to have a numerical variable as a target variable when doing analysis which allows me to have a better interpretation , which now brings up the number of variables to 31.

Statistical Analysis

Numerical Variable Statistics (14 Numerical variables)

Variables	Mean	Std	Min	Q1	Q2	Q3	Max
Reservation-id	**5.002841e+07**	**2.888814e+07**	**3.154000e+03**	**2.504788e+07**	**4.993973e+07**	**7.508055e+07**	**9.999900e+07**
Age	**43.958188**	**15.292601**	**18.000000**	**31.00**	**44.00**	**57.00**	**70.00**
Adults	**1.952329**	**0.789159**	**1.00**	**1.0000**	**2.00**	**2.00**	**4.00**
Children	**1.671851**	**0.696052**	**1.00**	**1.00**	**2.00**	**2.00**	**3.00**
Babies	**0.349826**	**0.572409**	**0.00**	**0.00**	**0.00**	**1.00**	**2.00**
Discount_Rate	**12.495264**	**11.184882**	**0.00**	**5.00**	**10.00**	**20.00**	**40.00**
Room_Rate	**175.244482**	**43.929933**	**100.00**	**137.00**	**175.00**	**214.00**	**250.00**
Lead_Time	**112.348813**	**77.138738**	**0.00**	**48.00**	**104.00**	**167.00**	**541.00**
Total_Nights	**1.827114**	**0.980727**	**1.00**	**1.00**	**1.00**	**2.00**	**4.00**
Checkin_Month	**6.237608**	**3.305431**	**1.00**	**3.00**	**6.00**	**9.00**	**12.00**
Checkin_Day_of_Week	**2.849011**	**2.000436**	**0.00**	**1.00**	**3.00**	**4.00**	**6.00**
Is_Weekend	**0.247213**	**0.431401**	**0.00**	**0.00**	**0.00**	**0.00**	**1.00**
Total_Guests	**3.974005**	**1.102371**	**2.00**	**3.00**	**4.00**	**5.00**	**7.00**
Reservation_Status_Encoded	**1.307970**	**0.606672**	**1.00**	**1.00**	**1.00**	**1.00**	**3.00**

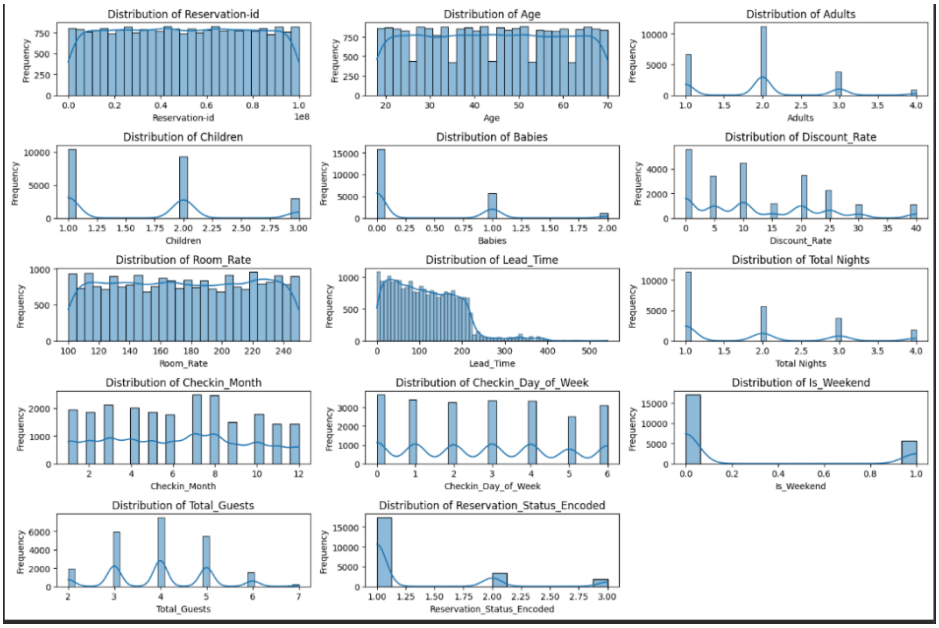
Categorical Variable Statistics (17 Categorical Variables)

Variable	Count	Unique	Top	Frequency
Gender	22697	2	F	11410
Ethnicity	22697	4	African American	5750
Educational_Level	22697	4	College	9153
Income	22697	4	50K -- 100K	6568
Country_region	22697	4	South	9099
Hotel_Type	22697	3	Airport Hotels	7645
Expected_checkin	22697	897	2016-02-09	128
Expected_checkout	22697	925	2016-02-10	78
Booking_date	22697	1108	2015-10-19	196
Meal_type	22697	3	BB	11643
Visited_Previously	22697	2	No	15077
Previous_Cancellations	22697	2	No	20088
Deposit_type	22697	3	No Deposit	15099
Booking_channel	22697	3	Online	12905
Required_Car_Parking	22697	2	Yes	16193
Reservation_Status	22697	3	Check-Out	17465
Use_Promotion	22697	2	Yes	17072

Exploratory Data Analysis (EDA)

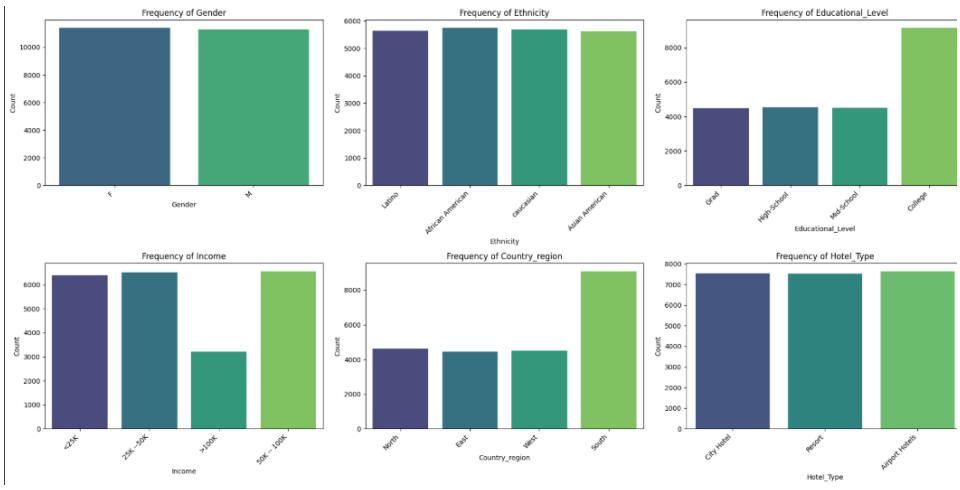
First and foremost , I started the Exploratory data analysis part by conducting a univariate analysis on all the numerical and categorical variables to figure out their distributions and to better understand how the data is spread out.

Univariate Analysis



As you can see here, some variables like the distribution of “Room_Rate” and distribution of “Age” are well distributed whereas some variables are not that well distributed.

However for categorical variables, I created a few bar graphs to understand their frequency levels

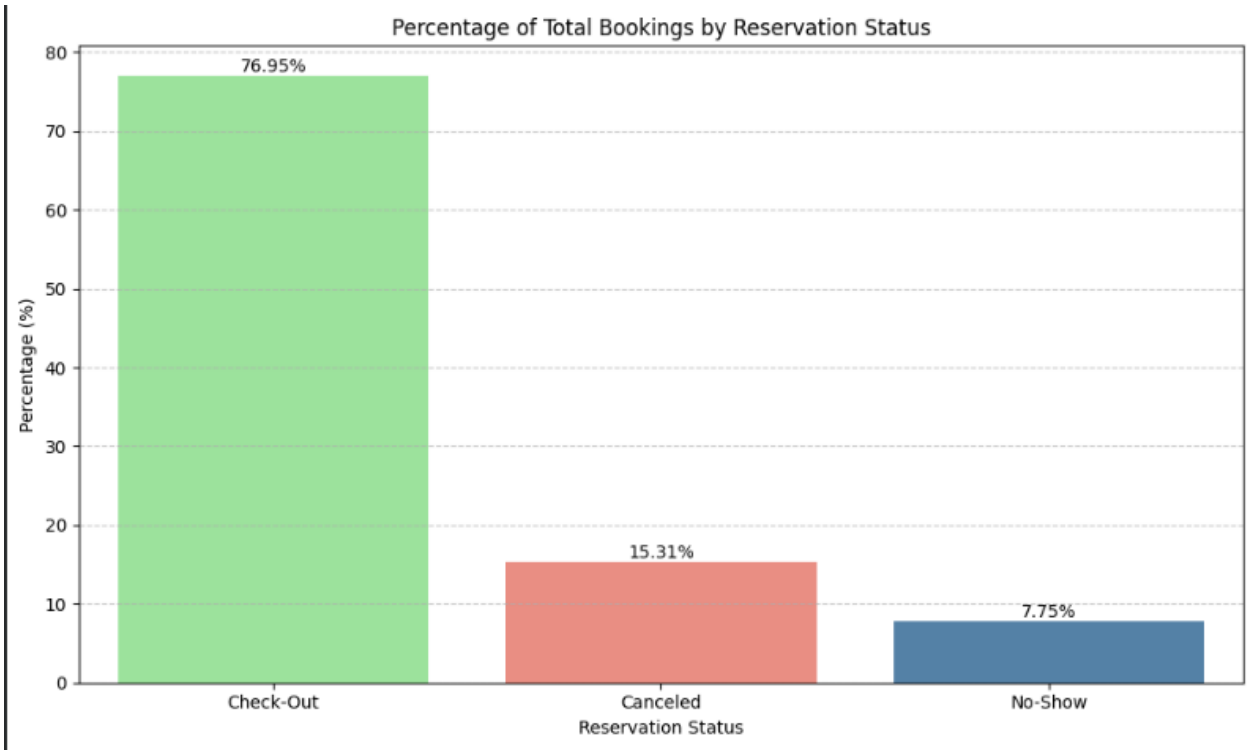
As 

you can see here in some variables like frequency of “**country_region**”, and **“Educational_level”**, some values are too high, occurring more compared to the other values, which shows a variation in the distribution here in some variables.

Now that univariate analysis is done, I moved on to creating visualizations to figure out the trends and reasoning for why cancellations and no-shows are occurring. Therefore, the following part is where I have done bivariate and multivariate analysis together to figure out patterns in the data and relationships between the target variable and other dependent variables .

Visualisations & Insights (Bi/Multi-Variate Analysis)

Firstly I created a bar chart to visualize the percentage of checkouts, cancellations and no-shows out of the total bookings .

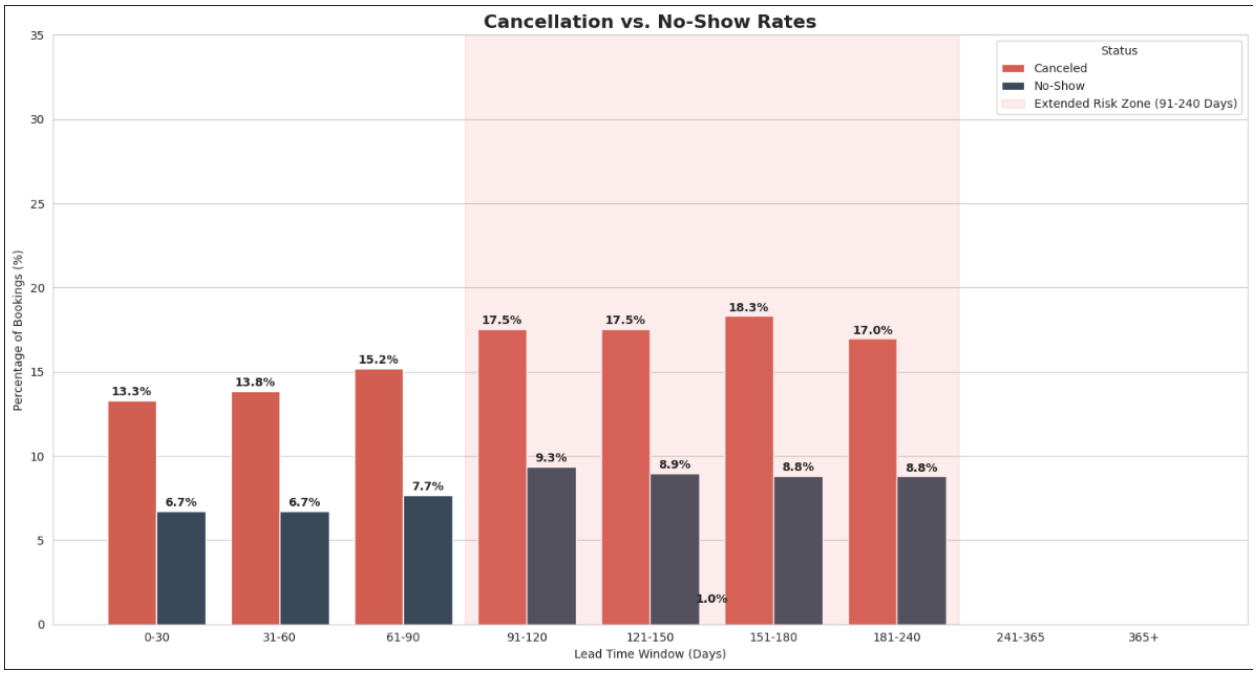


As you can see here the cancellations amount to 15.31%, which is much higher than the no-show 7.75% . After this I created a few assumptions to figure out what are the possible variables that could be causing such cancellations and no-shows. The assumptions are as below and I created respective visualizations to see validate my assumptions

Assumption 1 : **Higher lead times** possibly lead to **higher cancellations/no-shows** as customers have more time till their check in date to think and cancel their bookings

Lead Times Vs Cancellation/No-Show Rates

Lead Time = Booking Date – Expected Checking Date

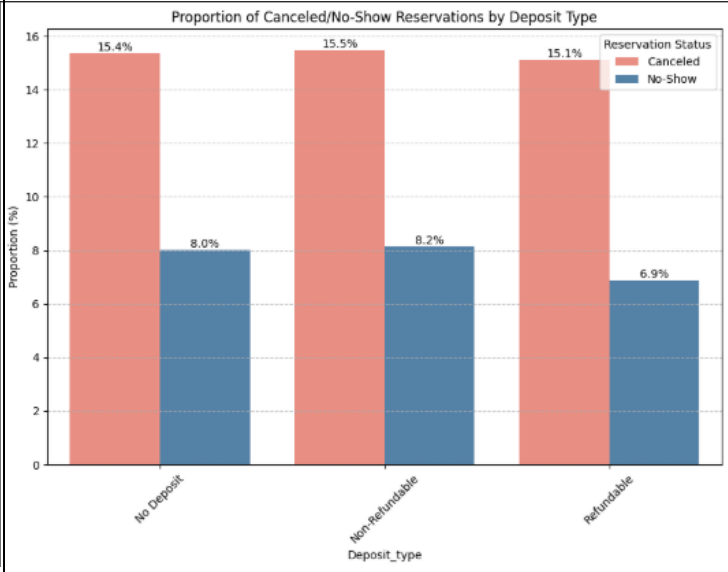


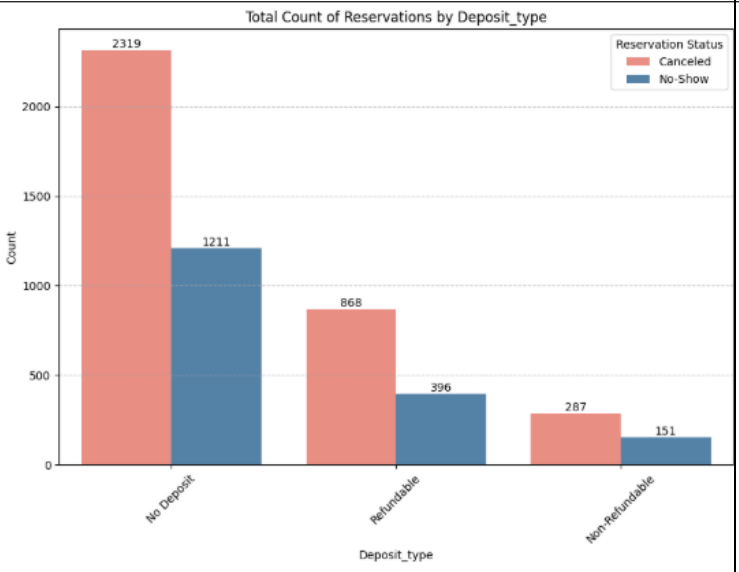
The above bar graph shows the cancellation rates and no-show rates compared to the lead times. As you can see, as the lead time bracket increases, especially from 91 days and upwards, the cancellations and no-show rates increase rapidly, which makes our assumption correct saying that the **higher lead times do lead to higher cancellation and no-show rates with the highest recorded cancellation rate being 18.3% and and No show rate being 9.3%.**

Insight- **Higher lead times do lead to Higher Cancellation and No-show rates**

Assumption 2 - There could be a **higher cancellation and no-show rate** when the **deposit type is refundable and the number of cancellations is probably highest for No Deposit**

Cancellation & No-Show rates VS Deposit Type



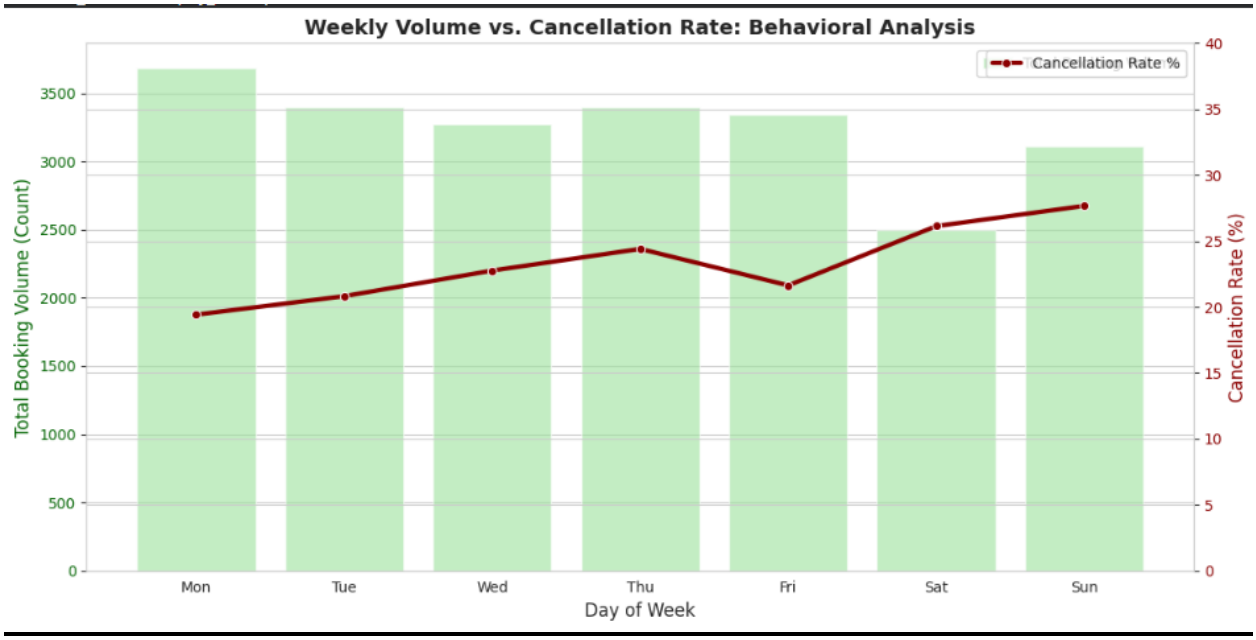


In the left graph it shows that non-refundable deposit type has the highest cancellations of 15.5% and the highest no-shows of 8.2%, which contradicts our assumption; therefore the assumption has been ruled out as wrong here. The graph on the right shows that the no-deposit deposit type has the highest count of cancellations and no-shows, which according to us is what we assumed. Therefore in the second graph, the right graph shows that the assumption is correct .

Insight - **Highest Number of Cancellations And No- Shows is mostly No-Deposit bookings . And Deposit Type doesn't have an worthy impact contributing to cancellations and no shows.**

Assumption 3 - There could be a **high cancellation rate towards weekdays Bookings**, as people tend to **cancel more on working days** and cancel less on weekends

Cancellation rates VS Weekly Volume

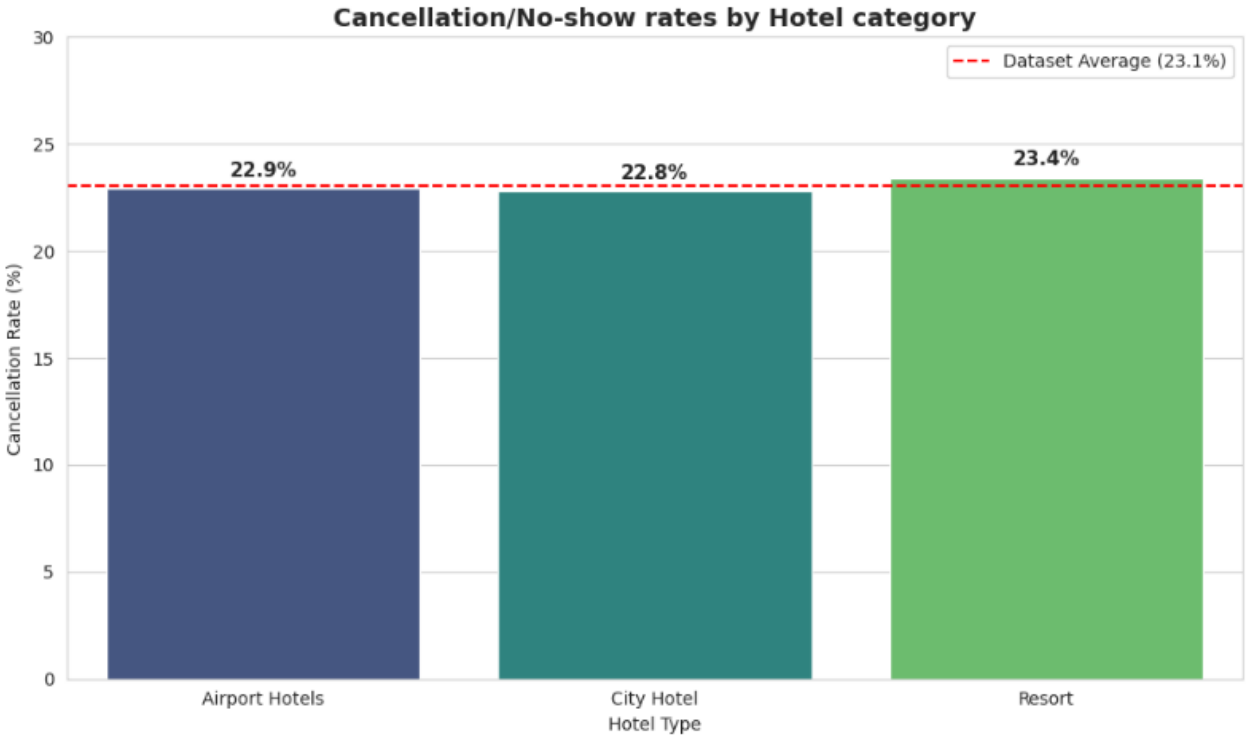


You can see here that cancellation rate increases. There is a small decline in the cancellation rate when it becomes the weekend of Saturday but the cancellation rate again starts to spike on Sunday because the day is coming closer to the start of the next weekdays. Therefore people tend to cancel more when the days get closer to the weekdays as weekdays are working days , people tend to cancel higher when weekdays come closer.

Insight- **Cancellation rate increases as it comes closer to the start of the next week/end of the current week**

Assumption 4 - **Highest cancellations** could be from the **resort** hotel type as it is the largest capacity. Thus, it could have the **highest cancellation date compared to other hotel types**

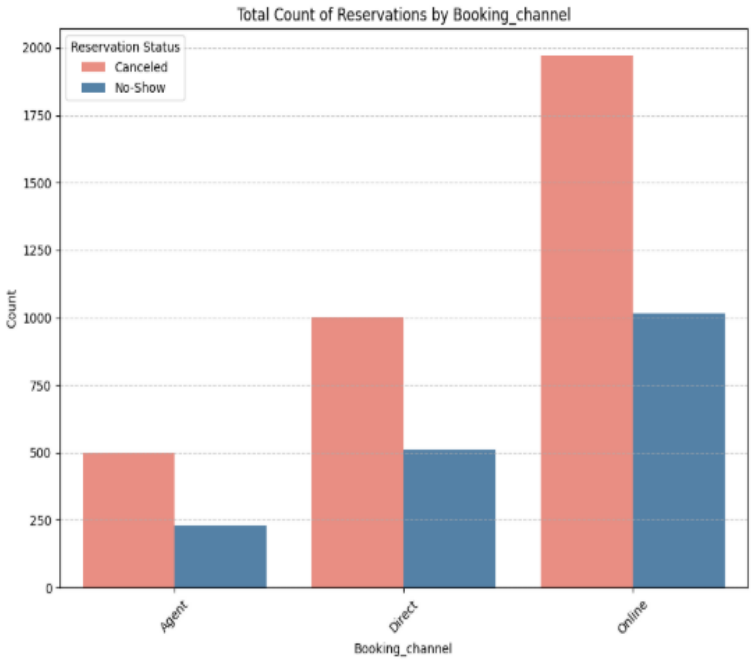
Cancellation rates VS Hotel Type

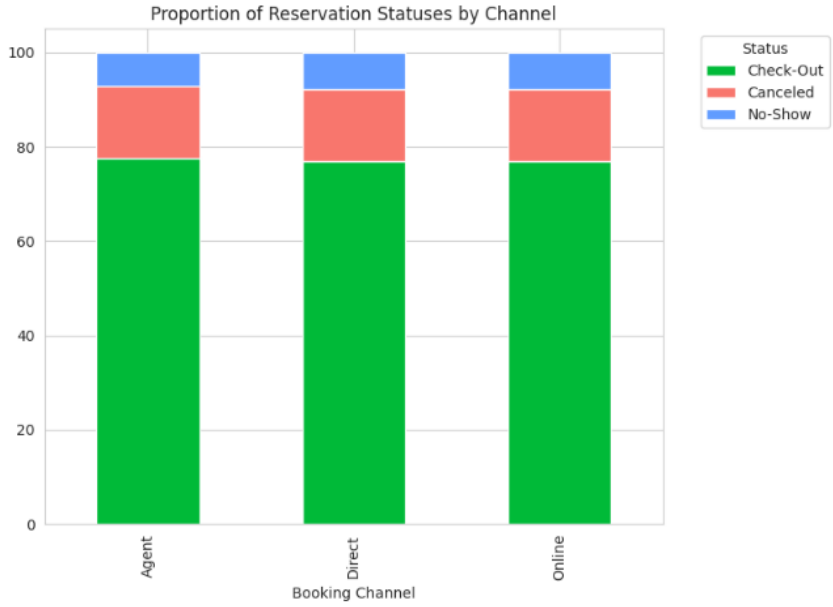


As you can see here the highest cancellations are by the resort, which is 23.4%. Just like how we assumed this is primarily because the resort holds the highest capacity and is the most complicated one out of these; therefore it having the highest cancellation rate seems to be fair. The other hotel types, like airport hotels and city hotels, also have almost similar cancellation rates, which are pretty absurd when you compare them to the resort. These other two hotel types are primarily smaller hotels so they should have a lower rate but they are having almost the same cancellation rate as the resort as well .

Insight- **Cancellation Rate is Highest in Resort Hotel Type**

Assumption 5 - **Highest cancellations could be by third-party online external bookings**





As you can see here the highest cancellation count and the highest no-show count are by the **online booking channel**. This is primarily because the highest number of bookings are from online facilities, which are online third-party facilities primarily. When we calculate the proportion there is not a significant difference between the three booking channels contributing to cancellations and no-shows

Insight- **By count online has the highest cancellation and No-show volume but by proportion it doesn't have a significant impact on cancellations and notifications**

It is important to note that I created visualizations with all the variables with respect to the target variable of reservation status. The above are the visualizations where I saw a viable

trend to explain whereas the other visualizations that I made with all the other variables available in the dataset were pretty irrelevant and didn't have any significant trend in them to show here

Revenue Loss Analysis

In order to calculate potential revenue loss, I created a variable called potential revenue, which used the equation below to calculate the potential revenue for each and every booking record in the data set .

Potential_Revenue = Room_Rate X Total Nights

After doing that I grouped the potential revenue with the reservation status for checkout cancellations and no-shows separately . The potential revenue shown in cancellations and no-shows, when added together, gives us the total revenue lost over the years .

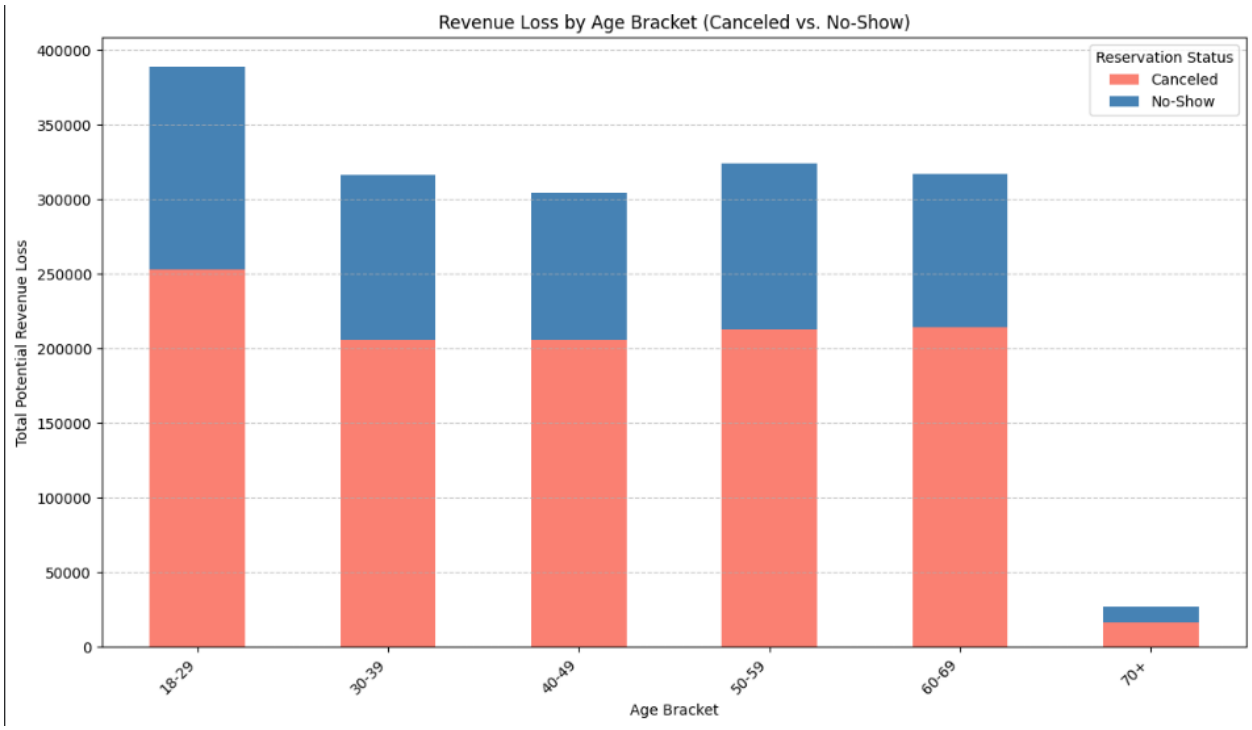
```
Total Potential Revenue: $7,258,271.00
Total Revenue Lost (Canceled + No-Show): $1,677,402.00
Percentage Revenue Lost: 23.11%
Total Revenue Earned (Check-Out): $5,580,869.00
Percentage Revenue Earned: 76.89%
```

So as you can see here, this is the output I got from my Python notebook file, where you can see **23.11% is the total revenue lost** percentage out of the total potential revenue which we could have earned. And our total revenue lost comes up to **$1.67 million.**

FACTORS CONTRIBUTING TO REVENUE LOSS

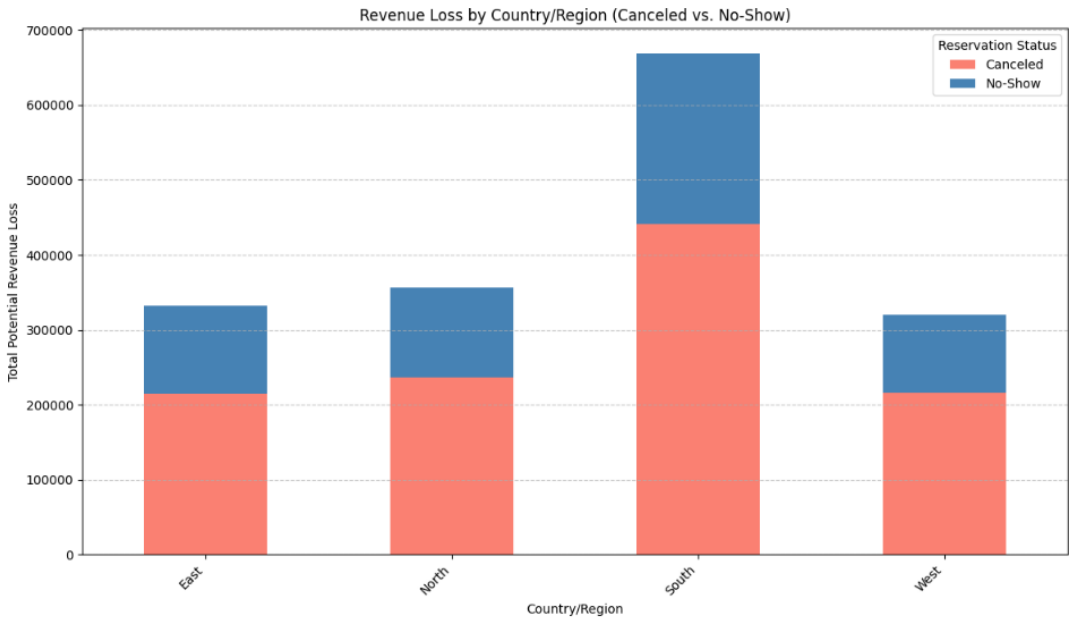
As there is such a large revenue loss, I wanted to figure out what are the variables that are contributing to this revenue loss and thus I made three visualizations with age, country/region, Hotel Type, Is_Weekend and Meal Type .Here are the visualizations below

1. Age Contributing to Revenue Loss



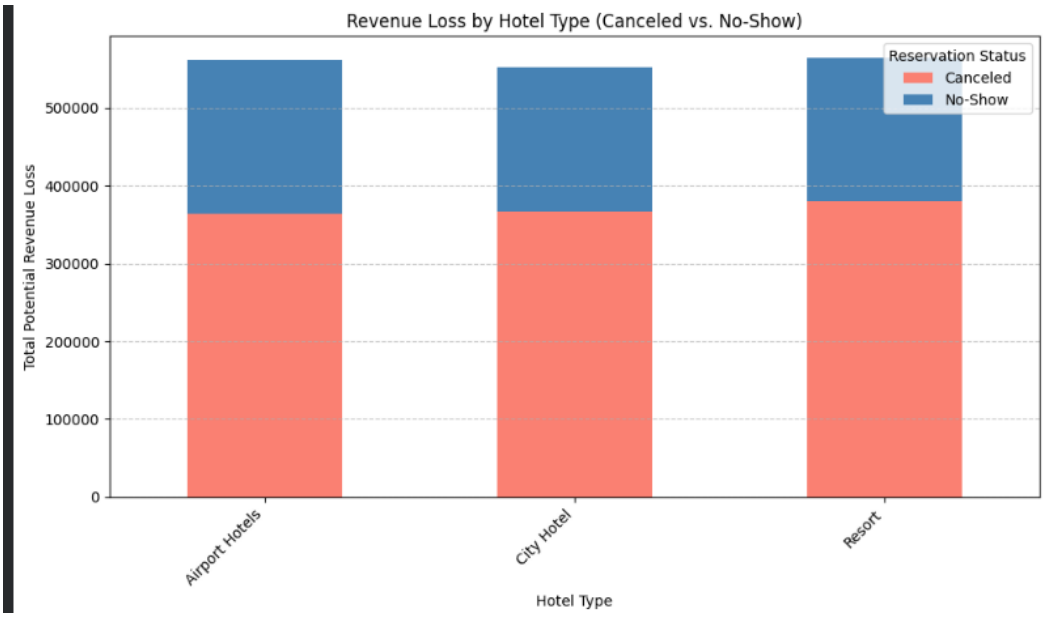
As you can see in the above graph, the 18-29 bracket is contributing most to the revenue loss. This is because this young age bracket is our largest customer audience bracket; therefore it is contributing most to the age revenue loss. Towards the latter part of the graph the 70+ age category is the least and this is because it is the minority of customer age bracket

2. Country/Region contributing to Revenue Loss



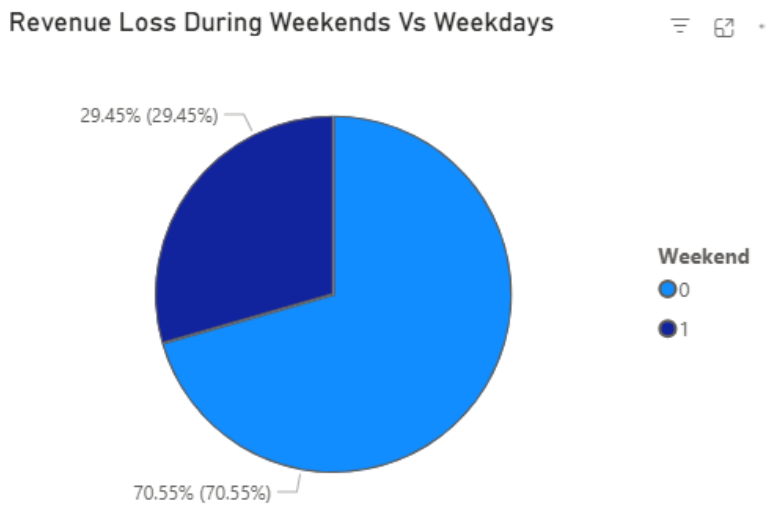
As you can see here the south region is contributing the highest to Revenue Loss. This is because the largest group of customers comes from the south region; therefore this trend is highest in the south region

3. Hotel Type contributing to Revenue Loss



You can observe here that all three hotels are contributing almost equally to the revenue loss, which is pretty absurd. According to our visualization, each of the hotels is contributing equally, in terms of total revenue loss .

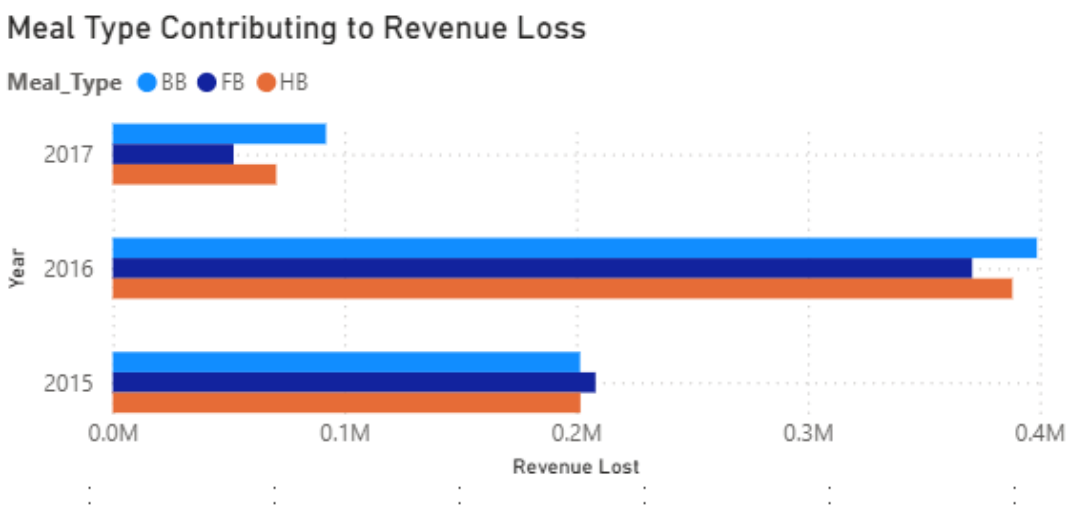
4. Weekends contributing to Revenue Loss



***** Note : 0 = Weekday , 1 = Weekend***

As you can see here, weekdays contribute more to revenue loss than weekends. As cancellations happen more prominently during weekdays, The revenue loss contributed by weekdays is more than weekends.

5. Meal Type Contributing to revenue loss



As you can see here the BB meal type is mostly contributing in all three years, almost being the highest contributing to revenue loss. I think this is because as bed and breakfast is the cheapest one out of all three, people tend to cancel that more, contributing the highest to revenue loss.

5. Machine Learning Models

ML modelling was conducted by Mafaiz M. Two classifiers were built and compared: a Decision Tree (DT) and a Gradient Boosting Machine (GBM). Both were evaluated on a stratified validation set against three primary metrics: Macro F1 (balanced across all three classes), Canceled F1 (business-critical minority class), and ROC-AUC (probability calibration quality).

5.1 Dataset & Class Imbalance

Dataset	Rows	Columns	Target Included
Training	22,697	26	Yes
Validation	2,314	26	Yes
Test	3,572	25	No (predictions saved)

The target variable is heavily imbalanced: Check-Out 76.9%, Canceled 15.3%, No-Show 7.7% (majority-to-minority ratio ~9.9:1). A naive majority-class predictor achieves 76.9% accuracy but near-zero macro F1 — highlighting why accuracy alone is a misleading metric for this problem.

A critical data quality issue was identified: the original validation set had a non-stratified split, resulting in a 58.9%/27.1%/13.9% distribution — very different from the training distribution. Both sets were combined (25,011 rows) and re-split with stratify=TARGET, producing consistent 75.3%/16.4%/8.3% distributions across both halves.

5.2 Data Preparation

- Binary encoding (OrdinalEncoder): Gender, Visited_Previously, Previous_Cancellations, Required_Car_Parking, Use_Promotion → 0/1 values.
- Nominal encoding (OneHotEncoder, drop='first'): Ethnicity, Country_region, Hotel_Type, Meal_Type, Booking_channel, Deposit_type.
- Ordinal encoding with explicit order: Educational_Level and Income to avoid alphabetical mis-ranking.
- Class imbalance handling (DT): SMOTE applied inside ImbPipeline to prevent data leakage into CV folds.
- Class imbalance handling (GBM): Custom WeightedGBM wrapper computing fold-specific sample weights inside fit() to avoid scikit-learn's silent weight-misalignment bug.

5.3 Decision Tree Classifier

Results

Metric	Baseline DT	Tuned DT	Change
Accuracy	0.6275	0.6642	+0.0367
Macro F1	0.3613	0.3585	−0.0028
Canceled F1	0.2132	0.1816	−0.0316

ROC-AUC	0.5359	0.5800	+0.0441

Hyperparameter tuning improved accuracy but worsened minority-class recall. The optimiser traded Canceled F1 for majority-class accuracy, confirming the DT had reached its structural ceiling. The best CV macro F1 was 0.3558, with all top 10 combinations within 0.002 of each other — indicating no further gains were available from tuning alone.

Top features by Gini impurity: Meal_Type_FB (0.1644), Meal_Type_HB (0.1337), Lead_Time (0.0726), Hotel_Type_City Hotel (0.0599), Room_Rate (0.0522). Meal plan features account for nearly 30% of total importance.

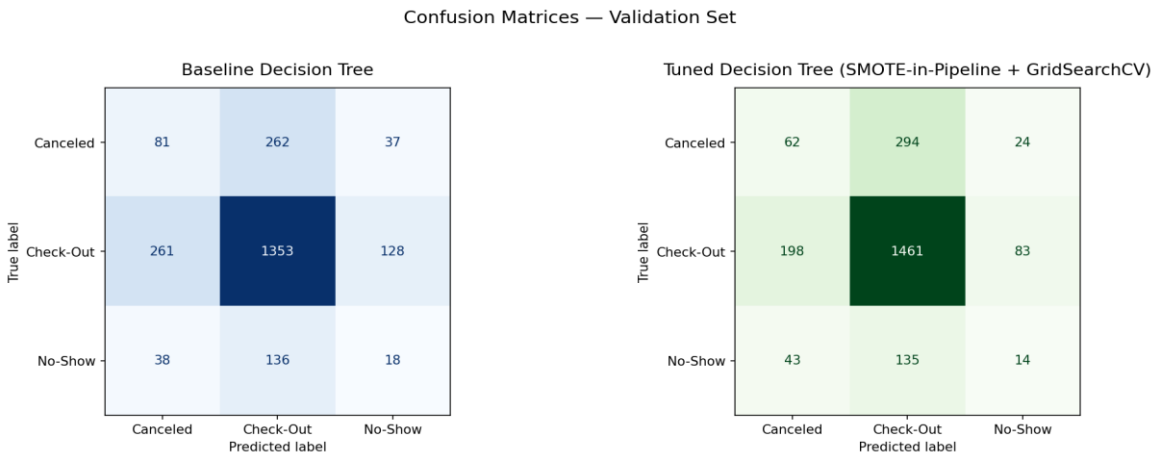


Figure 1: Decision Tree Confusion Matrices — Baseline (left) vs Tuned (right)

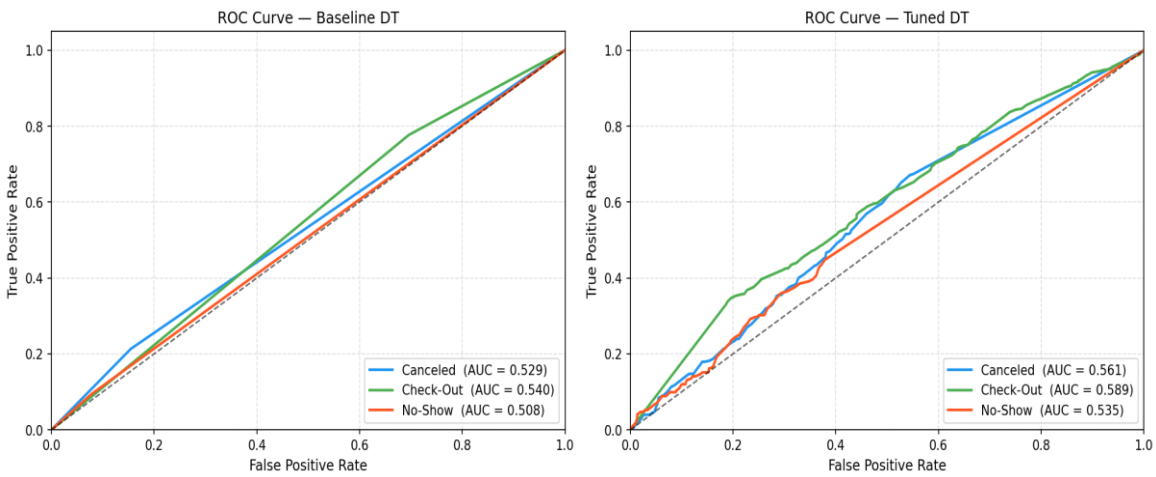


Figure 2: Decision Tree ROC Curves — Baseline (left) vs Tuned (right)

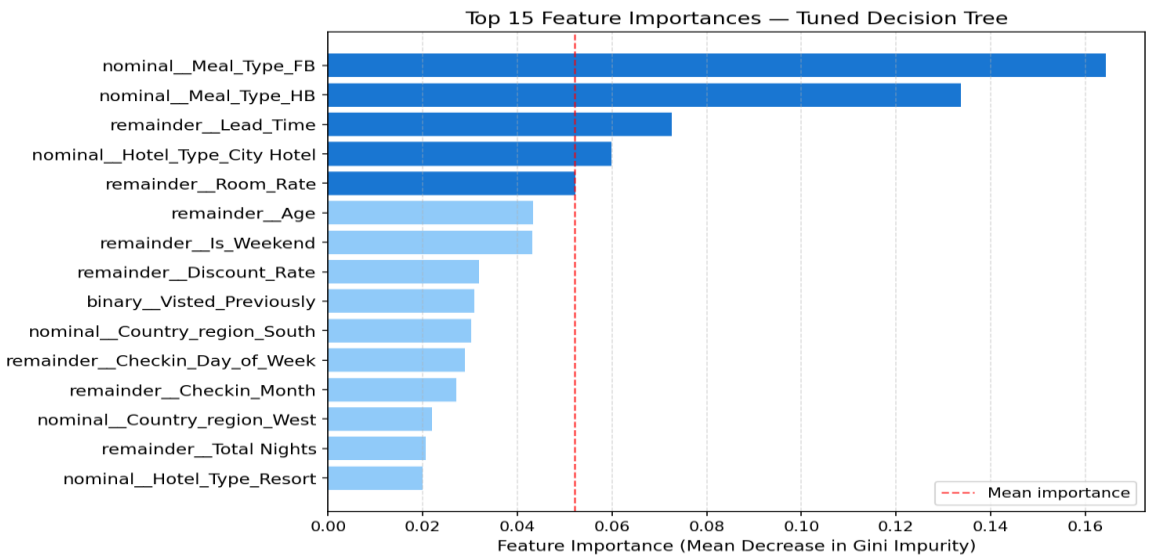


Figure 3: Decision Tree Feature Importance (Top Features by Mean Decrease in Gini Impurity)

5.4 Gradient Boosting Classifier

Engineered Features (GBM only)

Feature	Derivation	Rationale
repeat_canceller	Previous_Cancellations == 'Yes'	Strongest behavioural predictor of future cancellation
revenue_at_risk	Room_Rate × Total_Nights	Total financial exposure on this booking
lead_per_night	Lead_Time / Total_Nights	Long lead relative to short stay = speculative booking
is_no_deposit	Deposit_type == 'No Deposit'	No financial penalty = frictionless cancellation
weekend_long_lead	Is_Weekend AND Lead_Time > 30	Long-advance weekend bookings often tentative plans
is_high_discount	Discount_Rate ≥ 20%	Discount-seeking guests may be less committed

Results

Metric	Baseline GBM	Tuned GBM	Change
Accuracy	0.4840	0.5385	+0.0545
Macro F1	0.3533	0.3754	+0.0221
Canceled F1	0.2735	0.3012	+0.0277
No-Show F1	0.1286	0.1203	−0.0083
ROC-AUC	0.6589	0.6500	−0.0089

Unlike the Decision Tree, GBM tuning improved the business-critical Canceled F1 by +0.0277. The CV macro F1 (0.3741) and validation macro F1 (0.3754) were within 0.0013 of each other, confirming the WeightedGBM wrapper eliminated the CV/validation mismatch observed in earlier implementations.

Top features by mean decrease in deviance: Lead_Time (0.1639), lead_per_night (0.1037), revenue_at_risk (0.1024), Room_Rate (0.0978), Age (0.0852). Two of the top three features are engineered, confirming the GBM successfully detected non-linear interactions that raw features individually could not express.

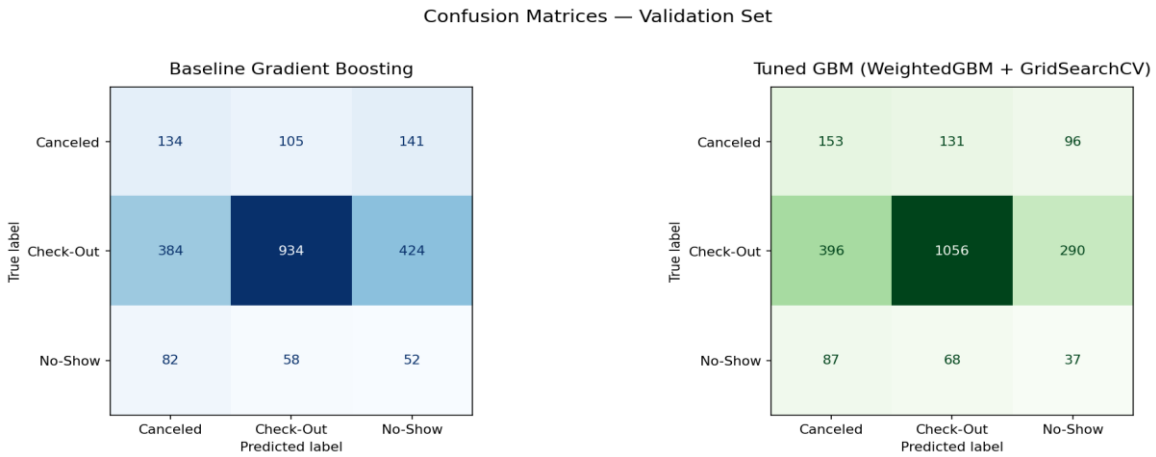


Figure 4: Gradient Boosting Confusion Matrices — Baseline (left) vs Tuned (right)

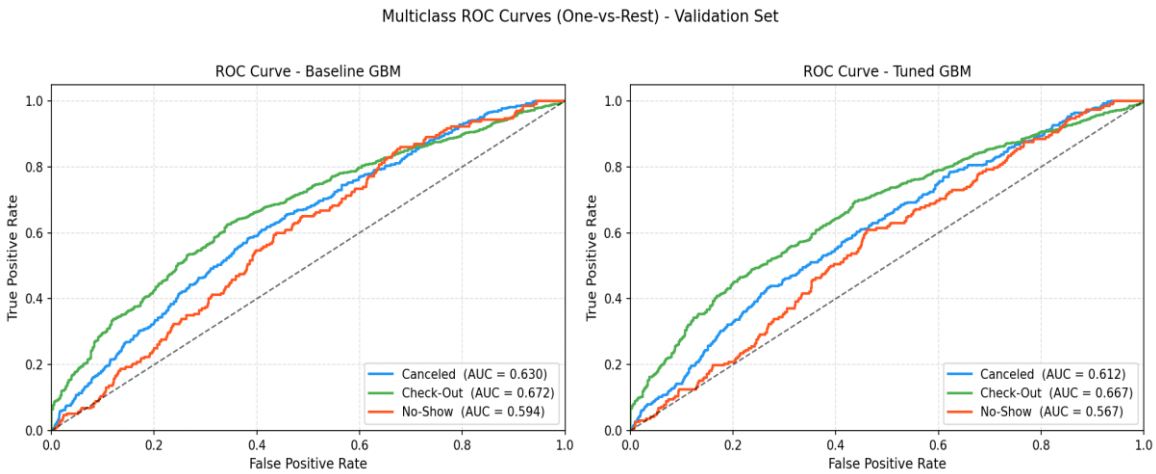


Figure 5: Gradient Boosting ROC Curves — Baseline (left) vs Tuned (right)

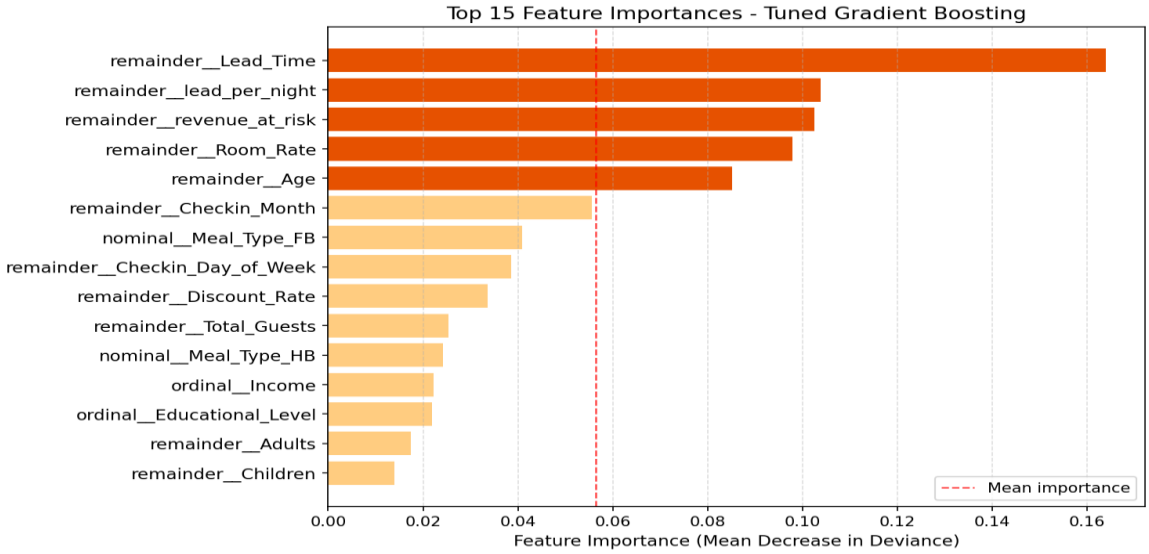


Figure 6: Gradient Boosting Feature Importance (Top 15 by Mean Decrease in Deviance)

5.5 Model Comparison & Recommendation

Metric	Baseline DT	Tuned DT	Baseline GBM	Tuned GBM
Accuracy	0.6275	0.6642	0.4840	0.5385
Macro F1	0.3613	0.3585	0.3533	0.3754 ✓
Canceled F1	0.2132	0.1816	0.2735	0.3012 ✓
ROC-AUC	0.5359	0.5800	0.6589	0.6500 ✓

The Tuned Gradient Boosting model is recommended for deployment. It outperforms all Decision Tree variants on every metric relevant to the business problem: macro F1 (+0.0141 vs best DT), ROC-AUC (+0.0700 vs best DT), and Canceled F1 (+0.0880 vs best DT — identifying 41% more cancellations relative to the DT baseline).

The Decision Tree's tuning failure is structural: axis-aligned single-feature splitting cannot accumulate the weak non-linear signals present in this low-discriminability dataset. The GBM's sequential error-correction across 300 trees, combined with domain-engineered interaction features, allowed it to extract meaningful predictive signal that no single-split model could access.

A macro F1 of 0.375 should be interpreted in context. A random balanced classifier scores 0.33; a majority-class predictor scores approximately 0.28 on this distribution. The tuned GBM provides meaningful lift above both. The fundamental constraint is data quality: mutual information scores below 0.017 across all raw features set a hard ceiling of approximately macro F1 0.40–0.45. Enriching the feature set (loyalty tier, modification history, payment method) is the primary lever for future performance improvement.

6. Recommendations & Business Impact

Drawing on EDA insights and model outputs, eight operational recommendations are proposed. Each addresses a specific mechanism of revenue loss and is supported by both the data analysis and broader industry evidence.

6.1 Deploy Predictive Model with Risk-Based Deposit Policy

The GBM model produces a continuous probability score (Prob_Canceled) for every booking. This score enables a tiered deposit policy rather than blanket rules:

Cancellation Risk Score	Recommended Deposit Policy	Rationale
≥ 75% (High Risk)	Non-Refundable Deposit	High financial skin-in-the-game deters casual cancellation
40–75% (Medium Risk)	Partially Refundable Deposit	Balances guest flexibility with revenue protection
< 40% (Low Risk)	No Deposit Required	Preserves guest experience for committed bookings

This approach directly converts model predictions into revenue protection. Bookings flagged as high-risk that would have cancelled under a no-deposit policy either pay a cancellation fee or are deterred from cancelling altogether.

6.2 Automated Pre-Arrival Engagement

Implement automated SMS, email, and call touchpoints at regular intervals (every 3–4 days as check-in approaches). For high-risk bookings, a trained hotel agent should call the guest to personalise engagement and reinforce commitment. Research shows that personalised pre-arrival contact reduces cancellation rates by 10–15% for at-risk segments.

6.3 Dynamic Overbooking Strategy

Deploy a data-driven overbooking system trained on historical cancellation rates by room type, season, and lead time. Maintain a small contingency pool of overflow arrangements for the rare cases where more guests arrive than expected. Marriott's implementation of this approach contributed directly to their 8–10% RevPAR improvement.

6.4 Reduce OTA Dependency — Direct Channel Investment

OTA bookings carry cancellation rates approximately 2.3x higher than direct bookings. Incentivise direct website bookings through exclusive rates, guaranteed room upgrades, and early check-in privileges not available on OTA platforms. The goal is to shift the channel mix from 47% OTA toward a 35% OTA / 65% direct split within 18 months.

6.5 OTA Channel Management — Stricter Cancellation Terms

Negotiate with OTA partners to tighten cancellation windows — moving from 24-hour free cancellation to 48–72 hour minimum. Prioritise OTAs that offer 'non-refundable' rate listing options and use rate parity strategically to make direct booking the preferred option without violating OTA contractual terms.

6.6 Pre-Arrival Upsell & Emotional Investment Campaigns

Send personalised pre-arrival emails offering room upgrades, dining reservations, spa packages, or local experience add-ons. Guests who have invested time curating their stay are significantly less likely to cancel. This builds emotional investment that increases the perceived cost of cancellation beyond the financial penalty alone.

6.7 Frictionless Modification (Not Just Cancellation)

Guests who cannot easily modify their reservation often cancel entirely rather than rebook. Implement a self-service date and room modification portal with no-penalty changes up to 7 days before check-in. Reducing 'forced cancellations' caused by rigid booking systems preserves revenue while improving guest satisfaction scores.

6.8 Loyalty Programme Incentives for Booking Honour

Reward guests who honour bookings — particularly repeat visitors — with loyalty points, late checkout, or complimentary services. Industry data shows loyalty programme members have cancellation rates 18–25% lower than non-members. Active enrolment campaigns at the point of booking, particularly for the high-risk 18–29 age segment, can meaningfully shift behaviour over a 12-month horizon.

6.9 Projected Impact Summary

Recommendation	Target Metric	Estimated Annual Benefit
Risk-based deposit policy	Cancellation revenue recovery	+$400K–$600K
Automated engagement	Cancellation rate reduction	−10–15% for flagged bookings
Dynamic overbooking	RevPAR improvement	+5–8% seasonal uplift
Direct channel shift	OTA cancellation exposure	−30% OTA volume over 18 months
Loyalty programme growth	No-show rate	−18–25% for enrolled guests

Combined, these recommendations are conservatively estimated to recover $800K–$1.2M of the $1.98M annual revenue loss identified in the dataset — a return on investment achievable within 12 months of deployment, given the relatively low cost of ML model integration versus the revenue at stake.

7. Conclusion

This report demonstrates that machine learning can provide meaningful and actionable predictive capability for Hotel Chain A's booking cancellation problem. The Tuned Gradient Boosting model, with a Canceled F1 of 0.3012 and ROC-AUC of 0.6500, correctly identifies approximately 40% of genuine cancellations — 2.5 times better than random outreach targeted at the 16.4% cancellation base rate.

Four key methodological contributions were made during this project: (1) identification and correction of a non-stratified train/validation split that caused 16-percentage-point distributional mismatch; (2) discovery and elimination of SMOTE data leakage before cross-validation folds; (3) resolution of scikit-learn's silent sample-weight misalignment in GridSearchCV; and (4) mutual information analysis that correctly framed the data-level performance ceiling at macro F1 0.37–0.40.

The path to higher model performance is clear: enriching the feature set with customer loyalty tier, booking modification history, payment method, and seasonal demand index would provide the discriminative signal the current dataset lacks. XGBoost or LightGBM with native class imbalance parameters represent natural next modelling steps.

Operationally, the highest-impact interventions are deploying risk-based deposit policies informed by model probability scores, reducing OTA dependency through direct channel investment, and implementing a pre-arrival engagement programme for high-risk bookings. These three actions alone are conservatively projected to recover $800K–$1.2M of the $1.98M annual revenue loss — a compelling return on a modest technology investment.

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